

Low Acrylonitrile NBR

POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Abstract

The patent landscape for Low Acrylonitrile NBR (Nitrile Butadiene Rubber) synthesis reveals a focus on emulsion polymerization processes. The majority of patents disclose the use of specific emulsifiers, initiators, and chain transfer agents to control the polymerization reaction. The acrylonitrile content in the resulting polymer is typically in the range of 18-46 wt%. Recurring chemistry includes the use of rosin acid or soap as emulsifiers, di-isopropylbenzene hydroperoxide as initiators, and tert-dodecyl mercaptan as chain transfer agents. Trends in the patent landscape indicate a focus on optimizing reaction conditions, such as temperature and reaction time, to achieve high conversion rates and desirable polymer properties. However, some parameters, such as monomer ratio and coagulation conditions, are not consistently disclosed across patents.

Methodology

Patent 1: Not disclosed

- Polymerization process: emulsion polymerization
- Acrylonitrile content: 18-46 wt%
- Monomer ratio: Not disclosed
- Water amount: Not disclosed
- Emulsifier: RA rosin acid or/and soap
- Initiator: di-isopropylbenzene hydroperoxide, iron edta sodium salt or secondary reduction agent rongalite
- Chain transfer agent: tert-dodecyl mercaptan, uncle's ten carbon mercaptan, uncle's 14 carbon mercaptan, uncle's 16 carbon mercaptan
- Temperature: 5-30 ■
- Reaction time: 7-12 hours
- Conversion: 65-80%
- Coagulation: Not disclosed
- Polymer properties: Mooney viscosity: 30-95, tensile strength: 10-22 Mpa
- Experimental notes: cold process discontinuous emulsion polymerization, nitrogen pressure testing, nitrogen replacement, cold water cooling, temperature-reducing coil

Cross-Patent Comparison & Synthesis Trends

Emulsifier Selection and Loading

The use of RA rosin acid or/and soap as emulsifiers is a common trend in the patent landscape.

Initiator and Chain Transfer Agent Strategies

Di-isopropylbenzene hydroperoxide is used as an initiator, while tert-dodecyl mercaptan and other mercaptans are used as chain transfer agents.

Monomer Ratio and Reaction Control

The monomer ratio is not disclosed in the provided patent, making it difficult to identify trends in this area.

Coagulation and Post-Treatment Conditions

Coagulation conditions are not disclosed in the provided patent, making it challenging to identify trends in this area.

References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year
Not disclosed	Not disclosed	Not disclosed	Not disclosed	Not disclosed